

■ DEPLOYMENT GUIDE

Complete Instructions: Download Models & Run Application

Multimodal Fake News Detection System CSE 499A Capstone Project Step-by-Step Setup & Deployment Guide

Quick Information:

Google Drive Link: <https://drive.google.com/drive/folders/1ZJAniZmCJB-nwnsM1VldWBPKlnMYOXUI>

Total Download Size: ~1.15 GB

Setup Time: ~30 minutes

Application: Streamlit Web App

Generated: August 11, 2026

1. System Requirements
2. Google Drive: What You'll Find
3. Phase 1: Initial Setup (5 minutes)
4. Phase 2: Download Models (10 minutes)
5. Phase 3: Application Setup (5 minutes)
6. Phase 4: Running the Application (3 minutes)
7. Phase 5: Testing & Verification (5 minutes)
8. Troubleshooting & FAQs
9. File Structure & Organization

Minimum Requirements

Component	Minimum	Recommended
Operating System	Windows 10/11, macOS, Linux	Windows 11 Pro
Python Version	3.10	3.10 or 3.11
RAM	8 GB	16 GB
Storage	5 GB free	10 GB free
GPU	Optional (CUDA 11.8+)	NVIDIA GPU with CUDA
Internet	Fast connection	1 Mbps+ for downloads

Software to Install

- ✓ Python 3.10+ (<https://www.python.org/downloads/>)
- ✓ Git (<https://git-scm.com/download/>)
- ✓ Google Drive access (or download client)
- ✓ Text Editor or IDE (VS Code, PyCharm, Jupyter)

Google Drive Folder Structure

Link: <https://drive.google.com/drive/folders/1ZJAniZmCJB-nwnsM1VldWBPklnMYOXUI>

- Fake_News_Dataset499A/
 - ■ CSE499A.ipynb (Training notebook)
 - ■ final_train_17k.tsv (Dataset: 17,045 samples)
 - ■ results/
 - ■■■ complete_results.json
 - ■■■ complete_results.csv
 - ■■■ complete_comparison.png
 - ■■■ xlm_roberta_trained.pt (1.04 GB) ■
 - ■■■ resnet_trained.pt (90 MB) ■
 - ■ images.zip (1.69 GB - Images archive)

Files You Need to Download

File	Size	Priority	Purpose
xlm_roberta_trained.pt	1.04 GB	■■■■ CRITICAL	Text model weights
resnet_trained.pt	90 MB	■■■■ CRITICAL	Image model weights
final_train_17k.tsv	4.5 MB	■ Optional	Dataset (for retraining)
images.zip	1.69 GB	■ Optional	Images (for retraining)

■■ IMPORTANT: You MUST download xlm_roberta_trained.pt and resnet_trained.pt. Without these files, the application will NOT work properly.

Step 1: Open Google Drive & Locate Folder

1. Open Google Drive: <https://drive.google.com>
2. Click the link: <https://drive.google.com/drive/folders/1ZJAniZmCJB-nwnsM1VldWBPKInMYOXUI>
3. You should see the folder structure
4. Verify you can access the 'results' subfolder
5. Open the 'results' folder

Step 2: Create Local Project Folder

Windows (PowerShell):

```
mkdir D:\CSE 499\FakeNewsApp  
cd D:\CSE 499\FakeNewsApp
```

macOS/Linux:

```
mkdir ~/FakeNewsApp  
cd ~/FakeNewsApp
```

Step 3: Verify Folder Created

You should now have an empty folder ready for:

- ✓ Model files (.pt files)
- ✓ Application code (app.py)
- ✓ Requirements file (requirements.txt)

Method A: Using Google Drive Web Interface

Step 1: Go to results Folder

1. Open: <https://drive.google.com/drive/folders/1ZJAniZmCJB-nwnsM1VldWBPKInMYOXUI>
2. Navigate to the 'results' subfolder

Step 2: Download xlm_roberta_trained.pt

1. Right-click on "xlm_roberta_trained.pt" (1.04 GB)
2. Click "Download"
3. Wait for download to complete (~3-5 minutes on fast internet)
4. Location: Downloads folder (usually)

Step 3: Download resnet_trained.pt

1. Right-click on "resnet_trained.pt" (90 MB)
2. Click "Download"
3. Wait for download (~30 seconds)

Step 4: Move Files to Project Folder

1. Open your Downloads folder
2. Find xlm_roberta_trained.pt
3. Cut (Ctrl+X) and paste into: D:\CSE 499\FakeNewsApp\
4. Do the same for resnet_trained.pt
5. Verify both files are in the folder

Method B: Using Google Drive Desktop Client (Faster)

1. Install Google Drive for Desktop: <https://support.google.com/drive/answer/7329379>
2. Sign in with your Google account
3. Navigate to Fake_News_Dataset499A > results
4. Right-click on xlm_roberta_trained.pt
5. Select "Make available offline"
6. Find the file in Google Drive folder on your computer
7. Copy to: D:\CSE 499\FakeNewsApp\
8. Repeat for resnet_trained.pt

✓ Verify Download Success:

Open File Explorer / Finder and check:

D:\CSE 499\FakeNewsApp\

■■■ xlm_roberta_trained.pt (1.04 GB)

■■■ resnet_trained.pt (90 MB)

If both files are present → YOU'RE ON TRACK! ■

Step 1: Create Python Virtual Environment

Windows (PowerShell):

```
cd D:\CSE 499\FakeNewsApp
python -m venv venv
.\venv\Scripts\Activate.ps1
```

macOS/Linux:

```
cd ~/FakeNewsApp
python3 -m venv venv
source venv/bin/activate
```

- ✓ You should see "(venv)" in your command line prompt
- ✓ This means the virtual environment is active

Step 2: Copy Application Files

Download the following files from the provided documentation:

1. **app.py** (Fixed version - app_FIXED_FINAL.py)
Source: app_FIXED_FINAL.py (rename to app.py)
Destination: D:\CSE 499\FakeNewsApp\

2. **requirements.txt**
Source: Create new file with dependencies
Destination: D:\CSE 499\FakeNewsApp\

■ requirements.txt should contain:

```
torch==2.0.1
torchvision==0.15.2
transformers==4.30.2
streamlit==1.27.0
numpy==1.24.0
pillow==9.5.0
requests==2.31.0
pandas==1.5.3
scikit-learn==1.2.2
```

Step 3: Install Dependencies

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

■■ **Wait Time:** This may take 5-10 minutes. The terminal will show package downloads. Do NOT close the terminal.

Step 1: Verify Setup

Before running the app, verify you have:

- ✓ Folder: D:\CSE 499\FakeNewsApp\
- ✓ File: app.py
- ✓ File: requirements.txt
- ✓ File: xlm_roberta_trained.pt (1.04 GB)
- ✓ File: resnet_trained.pt (90 MB)
- ✓ Virtual environment activated (venv)
- ✓ All dependencies installed

Step 2: Launch Streamlit Application

```
streamlit run app.py
```

Expected Output:

You can now view your Streamlit app in your browser.

Local URL: `http://localhost:8501`

Network URL: `http://192.168.x.x:8501`

■ Browser should open automatically to `http://localhost:8501`

Step 3: Check Model Loading

Wait for 10-15 seconds while the app loads models.

You should see in the web interface:

■ Text Model: ■ TRAINED WEIGHTS LOADED

■■ Image Model: ■ TRAINED WEIGHTS LOADED

If you see ■ for both → YOU'RE GOOD! ■

If you see ■ or warnings → Check troubleshooting section

Test 1: Upload Real News

1. In the Streamlit interface, find the text input area
2. Enter: "Scientists discover breakthrough in renewable energy technology"
3. Upload a genuine news image
4. Click "ANALYZE" button

Expected Result:

■ REAL NEWS (with 75%+ confidence)
Text Model: 80%+ (REAL)
Image Model: 70%+ (REAL)

Test 2: Upload Fake/Sensational News

1. Enter: "SHOCKING! This one WEIRD trick loses weight FAST! Doctors HATE this!"
2. Upload a low-quality or sensational image
3. Click "ANALYZE" button

Expected Result:

■■ FAKE NEWS (with 70%+ confidence)
Text Model: 75%+ (FAKE)
Image Model: 65%+ (FAKE)

If Both Tests Pass:

- CONGRATULATIONS! Your application is working perfectly!
- Models loaded correctly
- Text analysis working
- Image analysis working
- Fusion mechanism working
- UI responsive
- Ready for demonstration!

■ ***Problem: Model Files Not Found***

Error Message:

"FileNotFoundError: [Errno 2] No such file or directory: 'xlm_roberta_trained.pt'"

Solution:

1. Check file location: D:\CSE 499\FakeNewsApp\
2. Verify file names (no extra spaces or characters)
3. Re-download from Google Drive if corrupted
4. Use absolute path in app.py if needed

■ ***Problem: Module Not Found Error***

Error Message:

"ModuleNotFoundError: No module named 'transformers'"

Solution:

1. Activate virtual environment: .\env\Scripts\Activate
2. Reinstall packages: pip install -r requirements.txt
3. Use python -m pip if pip fails
4. Check Python version: python --version (should be 3.10+)

■ ***Problem: Models Show Base Model Warning***

Error:

Status shows "■■ BASE MODEL" instead of "■ TRAINED WEIGHTS"

Solution:

1. Models may be loading, wait 20-30 seconds
2. Check console for error messages
3. Verify file sizes match (1.04 GB and 90 MB)
4. Delete model files and re-download from Google Drive

■ ***Problem: Port 8501 Already in Use***

Error:

"Address already in use. Streamlit can't run."

Solution:

1. Another Streamlit instance is running
2. Kill the process: lsof -ti:8501 | xargs kill -9 (macOS)

3. Or use different port: `streamlit run app.py --server.port=8502`

Question	Answer
How long to download?	XLM (3-5 min), ResNet (30 sec) on fast internet
Do I need GPU?	No, CPU works but slower. GPU recommended
Can I use CPU only?	Yes, but inference ~10x slower
How long does inference take?	~1-2 seconds per prediction
Can I run without virtual env?	Not recommended, but possible

Final Project Directory Structure

```
D:\CSE 499\FakeNewsApp\  
├──  
│   ├── app.py ← Main application  
│   ├── requirements.txt ← Dependencies  
│   └── README_PROFESSIONAL.md ← Documentation  
├──  
│   ├── venv/ ← Virtual environment  
│   │   ├── Scripts/ ← Executables  
│   │   └── Lib/ ← Packages  
│   └──  
│       ├── xlm_roberta_trained.pt ← Text model (1.04 GB)  
│       └── resnet_trained.pt ← Image model (90 MB)  
  
Optional (for dataset retraining):  
├── final_train_17k.tsv ← Dataset  
└── images/ ← News images
```

File Size Summary

Component	Size
xlm_roberta_trained.pt	1.04 GB
resnet_trained.pt	90 MB
app.py	~15 KB
requirements.txt	~200 bytes
venv (virtual environment)	~500 MB
TOTAL PROJECT SIZE	~1.65 GB

All Commands in One Place

1■■■ CREATE FOLDER

Windows: `mkdir D:\CSE 499\FakeNewsApp`

macOS/Linux: `mkdir ~/FakeNewsApp`

2■■■ NAVIGATE TO FOLDER

Windows: `cd D:\CSE 499\FakeNewsApp`

macOS/Linux: `cd ~/FakeNewsApp`

3■■■ CREATE VIRTUAL ENVIRONMENT

All: `python -m venv venv`

4■■■ ACTIVATE VIRTUAL ENVIRONMENT

Windows: `.\venv\Scripts\Activate.ps1`

macOS/Linux: `source venv/bin/activate`

5■■■ INSTALL DEPENDENCIES

All: `pip install -r requirements.txt`

6■■■ RUN APPLICATION

All: `streamlit run app.py`

7■■■ DEACTIVATE ENVIRONMENT (when done)

All: `deactivate`

■ After Application is Running:

1. Test with real and fake news samples
2. Verify confidence scores are reasonable
3. Take screenshots of results for capstone
4. Note model loading time and performance
5. Test different input types (text length, image quality)

■ For Capstone Defense:

1. Keep application running during demo
2. Have sample real/fake news ready
3. Practice demo (3-5 minutes)
4. Be ready to explain architecture
5. Have this guide available for questions

■ Success Criteria:

- ✓ App runs without errors
- ✓ Models load (■ status shown)
- ✓ Real news scores high (80%+)
- ✓ Fake news detected (70%+)
- ✓ Inference time reasonable (~1.5 sec)
- ✓ UI responsive and professional

Generated: August 11, 2026 at 04:08 PM

For support, refer to README_PROFESSIONAL.md or troubleshooting section